Artificial Intelligence & Machine Learning – Task 5 Report

Ensemble Learning, Model Optimization & ML Pipeline

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1. Introduction

In this task, we explored advanced machine learning techniques focusing on **ensemble learning** and **model optimization**. The objective was to improve prediction performance by combining multiple models and tuning their parameters.

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2. Objective

- To implement different machine learning models
- To understand ensemble techniques like Random Forest and Gradient Boosting
- To optimize model performance using hyperparameter tuning
- To compare models and select the best one

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3. Dataset

We used the **Breast Cancer Dataset** available in sklearn.
This dataset contains features related to tumor characteristics and a target variable indicating whether the tumor is malignant or benign.

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4. Methodology

4.1 Data Preprocessing

- Loaded dataset using sklearn.datasets
- Split data into training and testing sets
- Applied feature scaling using StandardScaler (for Logistic Regression)

4.2 Models Implemented

1. Logistic Regression (Baseline Model)

- Used as a simple and interpretable model
- Provides a baseline for comparison

2. Random Forest

- Ensemble technique using multiple decision trees
- Reduces overfitting and improves accuracy

3. Gradient Boosting

- Sequential learning model
- Focuses on correcting previous errors

4.3 Hyperparameter Tuning

- Used **GridSearchCV** to find the best parameters
- Parameters tuned:
    - Number of estimators
    - Maximum depth

4.4 Model Evaluation

Models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-score

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5. Results

- Logistic Regression provided good baseline performance
- Random Forest improved accuracy due to ensemble learning

- Gradient Boosting showed strong performance with better error correction
- Tuned Random Forest (GridSearchCV) achieved the best performance

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6. Conclusion

Ensemble methods significantly improve model performance compared to single models. Hyperparameter tuning further enhances results. The final optimized model performed best and can be used for reliable predictions.

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7. Tools & Technologies Used

- Python
- Scikit-learn
- Pandas
- NumPy
- Jupyter Notebook

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8. Future Scope

- Implement advanced models like XGBoost and LightGBM
- Deploy the model using web frameworks
- Use real-world datasets for better generalization

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9. References

- Scikit-learn Documentation
- Kaggle Datasets
- Online tutorials and learning resources